

PREDICTION RESULTS

1. THE BASELINE SIMPLE LINEAR MODEL

Since this is the baseline model, we are establishing the "floor" for our performance.

- **R-Square (0.6146):** This model successfully explains **61.46%** of the variance in used car prices. For a first-pass linear model without polynomial curves or log transformations, this is a solid starting point.
- **Model Significance ($\text{Pr} > \text{F} < .0001$):** The overall model is statistically valid. It is definitively better than just guessing the average price of a car.
- **Root MSE (325,820):** Here is the model's weakness. With a Dependent Mean of 518,502, being off by an average of 325,820 means the model struggles heavily with pricing accuracy. This confirms what we suspected earlier: because you are predicting the *raw* selling price, a few super-expensive luxury cars are dragging the regression line up and inflating the error (Coefficient of Variation is nearly 63%).

Root MSE	325820	R-Square	0.6146
Dependent Mean	518502	Adj R-Sq	0.6134
Coeff Var	62.83884		

FEATURE IMPORTANCE & IMPACT

Identify important features:

- **Transmission (transmissionAutoma):** P-value < .0001, Estimate: +326,485. This is the single biggest value-add. Holding all else equal, an automatic transmission adds massive value to the car.
- **Engine Power (max_power_num):** P-value < .0001, Estimate: +318,677. Buyers pay a massive premium for horsepower.
- **Dealer Premium (seller_typeDealer):** P-value = 0.0032, Estimate: +196,694. Simply buying a car from a verified Dealer instead of an individual adds nearly 200k to the perceived value/price.

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	Intercept	1	685552	79703	8.60	<.0001
fuelCNG	fuel CNG	1	13015	52612	0.25	0.8046
fuelDiesel	fuel Diesel	1	50871	13372	3.80	0.0001
fuelLPG	fuel LPG	1	136046	59095	2.30	0.0214
seller_typeDealer	seller_type Dealer	1	196694	66798	2.94	0.0032
seller_typeIndividual	seller_type Individual	1	63095	65703	0.96	0.3369
transmissionAutoma	transmission Automa	1	326485	18254	17.89	<.0001
km_driven		1	8888.82989	17176	0.52	0.6048
seats		1	-12208	6127.72679	-1.99	0.0464
mileage_num		1	24376	7039.68625	3.46	0.0005
max_power_num		1	318677	8750.46176	36.42	<.0001
torque_value		1	5454.01489	4413.71289	1.24	0.2166
torque_rpm		1	-3226.79050	4387.87769	-0.74	0.4621
owner_encoded		1	-21403	5140.08459	-4.16	<.0001
car_age		1	-151054	7678.44438	-19.67	<.0001
km_per_year		1	-46019	16354	-2.81	0.0049
power_to_engine_ratio		1	-4772634	661393	-7.22	<.0001

THE VALUE DESTROYER :

- **Age (car_age):** P-value < .0001, Estimate: -151,054. The passage of time is the absolute harshest penalty on a car's price.
- **Usage Intensity (km_per_year):** P-value = 0.0049, Estimate: -46,019.
- **Ownership History (owner_encoded):** P-value < .0001, Estimate: -21,403. Every additional previous owner knocks roughly 21k off the resale value.

REAL-WORLD DECISION MAKING

If you were presenting this baseline model to a Dealership General Manager, here is how they would use it to make money tomorrow:

- **Smarter Purchasing:** The data proves that total km_driven doesn't inherently destroy value, but car_age and km_per_year do. The dealership should actively source older cars that have high total miles but were driven gently (low km_per_year). They can buy these cars cheap from individuals and sell them at a premium.

2. SIMPLE LASSO MODEL:

With LASSO, we are trading a tiny bit of accuracy for a massive gain in stability.

- **R-Square (0.6073) & Root MSE (328,662):** Your R-Square dropped by about 0.7%, and your error increased by about 2,800. *This is a good thing.* The Baseline model was "cheating" by memorizing statistical noise. LASSO penalized the model for using useless variables, giving you a slightly lower but much more honest accuracy metric.
- **The CV PRESS Criterion:** Look at the Selection Summary. The model stopped at **Step 9** because that is where the Cross-Validation (CV) Error reached its absolute minimum (5.877E14). This proves LASSO successfully stopped the model from destroying itself with too many variables.

Root MSE	328662
Dependent Mean	518502
R-Square	0.6073
Adj R-Sq	0.6067

Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Value
Model	9	9.226523E14	1.025169E14	949.07
Error	5523	5.96587E14	1.080187E11	
Corrected Total	5532	1.519239E15		

FEATURE IMPORTANCE & IMPACT

LASSO's primary job is feature selection—it physically crushes the coefficients of useless variables down to zero.

- **The Casualties (What LASSO killed):**
- LASSO completely eliminated seats, mileage_num, torque_value, torque_rpm, and specific fuel types like LPG and CNG.
- Look at engine_num in your Parameter Estimates: **The Estimate is 0.** LASSO mathematically neutralized the engine size because max_power_num already provides that information. Keeping both would cause multicollinearity, so LASSO removed one.

Parameter Estimates		
Parameter	DF	Estimate
Intercept	1	448909
fuelDiesel	1	67108
seller_typeDealer	1	111839
transmissionAutoma	1	306870
km_driven	1	-22486
engine_num	1	0
max_power_num	1	268287
owner_encoded	1	-15069
car_age	1	-133301
power_to_engine_rati	1	-63996

LASSO Selection Summary				
Step	Effect Entered	Effect Removed	Number Effects In	CV PRESS
0	Intercept		1	1.51951E15
1	max_power_num		2	8.17338E14
2	car_age		3	6.62272E14
3	transmissionAutoma		4	6.22688E14
4	seller_typeDealer		5	6.13788E14
5	fuelDiesel		6	6.0773E14
6	owner_encoded		7	6.05456E14
7	km_driven		8	6.033E14
8	engine_num		9	6.04311E14
9	power_to_engine_rati		10	5.8776E14*
10		engine_num	9	6.01412E14
11	km_per_year		10	6.0406E14
12	fuelLPG		11	6.03493E14
13	mileage_num		12	6.0392E14
14	torque_value		13	6.03822E14
15	seats		14	6.03643E14
16	engine_num		15	5.90189E14
17		seats	14	5.9028E14
18	torque_rpm		15	1.17046E18
19		km_driven	14	1.1237E18
20	seats		15	1.11999E18
21	km_driven		16	1.16619E18
22	fuelCNG		17	1.17098E18
23	seller_typeIndividua		18	1.17076E18
* Optimal Value of Criterion				

THE SURVIVORS (THE TRUE DRIVERS OF VALUE):

- **Transmission (transmissionAutoma):** Estimate: +306,870. Still the undisputed king of adding value.
- **Power (max_power_num):** Estimate: +268,287. Replaces all other mechanical stats (engine size, torque) as the single source of truth for mechanical value.
- **Age (car_age):** Estimate: -133,301. Remains the strongest factor for depreciation.
- **Dealer Premium (seller_typeDealer):** Estimate: +111,839. Notice LASSO kept the Dealer premium but dropped the Individual seller type. You only need to know if it's a dealer to see the price bump.

Parameter Estimates		
Parameter	DF	Estimate
Intercept	1	448909
fuelDiesel	1	67108
seller_typeDealer	1	111839
transmissionAutoma	1	306870
km_driven	1	-22488
engine_num	1	0
max_power_num	1	268287
owner_encoded	1	-15089
car_age	1	-133301
power_to_engine_rati	1	-63998

REAL-WORLD DECISION MAKING

If you are handing this model over to the Dealership's IT and Sales teams, here is how it changes the business:

- **Massive Reduction in Data Entry (Time = Money):** The Intake Team no longer needs to inspect or record torque, exact engine CCs, the number of seats, or minor fuel types. The model proves they only need **9 key data points** to price a car accurately. This speeds up the intake process and drastically reduces human data-entry errors.
- **Stable Deployment:** Because this model was optimized using Cross-Validation, the IT team can safely deploy it to a live pricing website. The Baseline model would have wildly mispriced a car if it had a weird "Torque" value; the LASSO model will ignore the weird torque and give a stable, safe price based on Power and Age.
- **The Diesel Pivot:** LASSO kept fuelDiesel (+67,108) but dropped the others. The dealership should specifically filter their auction lists for Automatic, Diesel vehicles with high Power to maximize their profit margins.

3. POLYNOMIAL LASSO REGRESSION

- Model Evaluation (The Breakthrough)
- This model absolutely destroyed the baseline.
- **R-Square (0.8388)**: You jumped from explaining ~61% of price variance to **83.88%**. That is a massive 20-point leap. You are now capturing the vast majority of the "why" behind a car's price.
- **Root MSE (217,551)**: Your average pricing error dropped from ~328,000 down to 217,000. You have reduced the dealership's financial risk by over 100,000 per vehicle priced!
- **The CV PRESS & ASE**: The model used Cross Validation (5-folds) to ensure this massive R-Square isn't just overfitting. The fact that LASSO kept 38 complex terms but stopped there, means it found true, generalizable patterns in the data.

Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Value
Model	55	1.267458E15	2.304485E13	501.29
Error	5477	2.517838E14	45971106782	
Corrected Total	5532	1.519239E15		

Root MSE	214409
Dependent Mean	518502
R-Square	0.8343
Adj R-Sq	0.8326
AIC	141433
AICC	141434
BIC	135888
C(p)	757.35417
SBC	136269
ASE	45505828998
CV PRESS	2.687994E14

2. FEATURE IMPORTANCE & IMPACT

car_age (-175,434) & car_age^2 (+16,335): This is the **Depreciation Curve**. The negative base term means the car loses massive value early on.

- **max_power_num^2 (+36,396):** Horsepower doesn't scale linearly. The higher the horsepower gets, the exponentially more expensive the car becomes.
- **transmissionAutoma * max_power_num (+103,657):** This is brilliant. The model found that having high horsepower is good, but having high horsepower *combined* with an Automatic transmission commands a massive, specific premium.
- **max_power_num * car_age (-117,978):** High horsepower is only valuable on *new* cars. If an old car has high horsepower, buyers assume it's a maintenance nightmare, and the value tanks.
- **fuelDiesel * transmissionAutoma (+203,687):** The ultimate combination for resale value. Buyers will pay a heavy premium specifically for Automatic Diesels.
- **transmissionAutoma * km_per_year (-148,210):** While Automatics are highly valued, if an Automatic car is driven heavily (high km per year), buyers heavily penalize the price, likely fearing an expensive transmission failure.

REAL-WORLD DECISION MAKING

This model elevates the dealership from "guessing" to "surgical precision." Here is the business strategy:

- **The Sourcing Matrix:** Sourcing agents should immediately be handed a specific checklist. They should be hunting exclusively for **Automatic Diesels** (+203k interaction). When sourcing high-horsepower vehicles, they must strictly avoid older models (-117k penalty) and only buy them young.
- **The Mileage Pivot:** The model shows that older cars with high total miles don't lose as much value if the km_per_year is low ($\text{car_age} * \text{km_per_year} = +9,122$). The dealership can buy older, well-maintained cars cheaply and sell them reliably.
- **Dynamic Pricing Engine:** Because the RMSE dropped by over 100k, this math should be hard-coded into the Dealership's internal dashboard. When a salesperson takes a trade-in, the system will apply the Age^2 curve to ensure they aren't offering too much money for a car that is in its steepest phase of depreciation.

MULTINOMIAL LOGISTIC REGRESSION

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	6191.6766	34	<.0001
Score	4294.2952	34	<.0001
Wald	1659.7812	34	<.0001

Model Evaluation (The "Too Perfect" Warning)

- This model has strong predictive power, but it is currently mathematically unstable.
- Global Null Hypothesis ($Pr > ChiSq < .0001$):** The model works. It is significantly better than random guessing at putting cars into the correct price tier.
- The Symptom:** Look at your Odds Ratios for torque_rpm, seller_type, and power_to_engine_ratio. They say >999.999 or <0.001. The model panicked and pushed the multipliers to infinity. In a production environment, you would fix this by dropping the offending variables or grouping your categories differently.

Odds Ratio Estimates				
Effect	price_category_encoded	Point Estimate	95% Wald Confidence Limits	
fuelCNG	1	1.250	0.447	3.491
fuelCNG	2	<0.001	<0.001	>999.999
fuelDiesel	1	2.385	1.623	3.505
fuelDiesel	2	81.754	43.250	154.535
fuelLPG	1	3.303	1.208	9.030
fuelLPG	2	0.007	<0.001	>999.999
seller_typeDealer	1	<0.001	<0.001	>999.999
seller_typeDealer	2	<0.001	<0.001	>999.999
seller_typeIndividual	1	<0.001	<0.001	>999.999
seller_typeIndividual	2	<0.001	<0.001	>999.999
transmissionAutoma	1	2.885	1.152	7.225
transmissionAutoma	2	7.646	2.855	20.482
km_driven	1	2.914	1.768	4.802
km_driven	2	6.364	2.986	13.564
seats	1	1.105	0.914	1.334
seats	2	1.327	1.051	1.676
mileage_num	1	1.389	1.124	1.717
mileage_num	2	1.267	0.967	1.661
engine_num	1	18.376	9.721	34.735
engine_num	2	136.860	58.862	318.212
max_power_num	1	0.405	0.189	0.871
max_power_num	2	0.405	0.155	1.055
torque_value	1	1.076	0.993	1.166
torque_value	2	1.186	0.766	1.837
torque_rpm	1	14.987	<0.001	>999.999
torque_rpm	2	>999.999	>999.999	>999.999
owner_encoded	1	0.781	0.701	0.870
owner_encoded	2	0.575	0.471	0.703
car_age	1	0.044	0.032	0.061
car_age	2	0.001	<0.001	0.002
km_per_year	1	0.293	0.173	0.495
km_per_year	2	0.139	0.069	0.279
power_to_engine_rati	1	>999.999	>999.999	>999.999
power_to_engine_rati	2	>999.999	>999.999	>999.999

FEATURE IMPORTANCE & IMPACT

Because the target is price_category_encoded (Tier 0 = Cheap, Tier 1 = Average, Tier 2 = Expensive), we evaluate impact using **Odds Ratios (OR)**.

OR > 1: Increases the odds of moving to a higher price tier.

OR < 1: Decreases the odds (pushes the car toward Tier 0).

- Let's look at the healthy variables predicting **Tier 2 vs. Tier 0**:
- Diesel Engine (fuelDiesel2):** OR = **81.75**. This is staggering. Holding all else equal, a Diesel car is *81 times more likely* to be in the Expensive tier compared to the baseline fuel type.

Odds Ratio Estimates		
Effect	price_category_encoded	Point Estimate
fuelCNG	1	1.250
fuelCNG	2	<0.001
fuelDiesel	1	2.385
fuelDiesel	2	81.754
fuelLPG	1	3.303
fuelLPG	2	0.007
seller_typeDealer	1	<0.001
seller_typeDealer	2	<0.001
seller_typeIndividua	1	<0.001
seller_typeIndividua	2	<0.001
transmissionAutoma	1	2.885
transmissionAutoma	2	7.646
km_driven	1	2.914
km_driven	2	6.364
seats	1	1.105
seats	2	1.327
mileage_num	1	1.389
mileage_num	2	1.267
engine_num	1	18.376
engine_num	2	138.860
max_power_num	1	0.405
max_power_num	2	0.405
torque_value	1	1.076
torque_value	2	1.186
torque_rpm	1	14.987
torque_rpm	2	>999.999
owner_encoded	1	0.781
owner_encoded	2	0.575
car_age	1	0.044
car_age	2	0.001
km_per_year	1	0.293
km_per_year	2	0.139
power_to_engine_rati	1	>999.999
power_to_engine_rati	2	>999.999

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- **Transmission (transmissionAutoma2):** OR = **7.64**. An automatic car is 7.6 times more likely to be categorized as Expensive.
 - **Age (car_age2):** OR = **0.001**. For every single year a car ages, its odds of being in the Expensive tier multiply by a tiny fraction. Age absolutely crushes a vehicle's chances of being top-tier.
 - **Ownership History (owner_encoded2):** OR = **0.575**. Every time a car changes owners, its odds of staying in the Expensive tier are cut nearly in half.

REAL-WORLD DECISION MAKING

- Why would a dealership use this tier-based Logistic model instead of the exact-price Polynomial model from Phase 3? Because business operates in buckets.
- **The "Fast Lane" Intake:** When a customer brings a car in for a trade-in, the Intake Desk doesn't need to calculate the exact dollar amount. They just punch in the Age, Transmission, and Fuel Type. If the model flags it as "Tier 2 (Expensive)," it gets immediately routed to a Senior Appraiser. If it flags as "Tier 0 (Cheap)," it goes to the standard lot.
- **Inventory Balancing:** The dealership's marketing team can use this to quickly categorize thousands of cars at auction. Instead of looking at precise prices, they can filter the auction list to show *only* cars that have a >80% probability of being "Tier 2" based on their specs, allowing them to stock up on premium inventory rapidly.